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E-Commerce Order Management System

Problem Statement:
Ensure customer orders are processed correctly with accurate inventory updates and payment handling.

Tasks:

1. Develop order processing and inventory management logic
2. Write JUnit tests for order success/failure scenarios
3. Implement CI/CD pipeline with automated testing
   - Set up version control using GitHub
   - Configure CI pipeline (Jenkins / GitHub Actions)
   - Automate build process using Apache Maven
   - Integrate automated unit tests in pipeline
   - Configure CD pipeline for deployment
4. Deploy services on Docker and Kubernetes
   - Containerize application using Docker
   - Push image to Docker Hub
   - Create Kubernetes deployment and service YAML files

Components	Marks
Java Program Development	15
Junit Testing	15
CI/CD Pipeline Creation and Execution	10
Deploy services on Docker and Kubernetes	10
Total:	**50**

File Edit Selection View Go Run Terminal Help

agile50

EXPLORER

AGILE50

- .github
- docker
  - docker-compose.yml
- k8s
  - deployment.yaml 1
  - service.yaml
- src
  - main
  - test\java\com\agile50\ecommerce\s...
    - OrderServiceTest.java
- target
  - classes\com\agile50\ecommerce
    - exception
    - model
    - payment
    - service
    - App.class
  - generated-sources\annotations
  - generated-test-sources\test-annotations
  - maven-status
  - surefire-reports
  - test-classes\com\agile50\ecommerce...
    - OrderServiceTest.class
- .gitignore
- Dockerfile
- Jenkinsfile M
- pom.xml
- README.md

OUTLINE

TIMELINE

JAVA PROJECTS

deployment.yaml 1 | Jenkinsfile M

BLACKBOX

Jenkinsfile

```
1   pipeline {
15      stages {
54          stage( Deploy to Kubernetes ) {
69              }
70          }
71
72          post {
73              always {
74                  junit testResults: 'target/surefire-reports/*.xml', allowEmptyResults: true
75              }
76          }
77      }
```

PROBLEMS 4 | OUTPUT | DEBUG CONSOLE | TERMINAL | PORTS | QUERY RESULTS

```
PS C:\Users\Jyothiswar Reddy\OneDrive\Desktop\agile50> git init
Reinitialized existing Git repository in C:/Users/Jyothiswar Reddy/OneDrive/Desktop/agile50/.git/
PS C:\Users\Jyothiswar Reddy\OneDrive\Desktop\agile50> git add .
PS C:\Users\Jyothiswar Reddy\OneDrive\Desktop\agile50> git commit -m "Initial"
[main (root-commit) 3a80cfb] Initial
 21 files changed, 623 insertions(+)
 create mode 100644 .github/workflows/cd.yml
 create mode 100644 .github/workflows/ci.yml
 create mode 100644 .gitignore
 create mode 100644 Dockerfile
 create mode 100644 Jenkinsfile
 create mode 100644 README.md
 create mode 100644 docker/docker-compose.yml
 create mode 100644 k8s/deployment.yaml
 create mode 100644 k8s/service.yaml
 create mode 100644 pom.xml
 create mode 100644 src/main/java/com/agile50/ecommerce/App.java
 create mode 100644 src/main/java/com/agile50/ecommerce/exception/InsufficientInventoryException.java
 create mode 100644 src/main/java/com/agile50/ecommerce/exception/PaymentFailedException.java
 create mode 100644 src/main/java/com/agile50/ecommerce/model/Order.java
 create mode 100644 src/main/java/com/agile50/ecommerce/model/OrderItem.java
 create mode 100644 src/main/java/com/agile50/ecommerce/model/OrderStatus.java
 create mode 100644 src/main/java/com/agile50/ecommerce/model/Product.java
```

File Edit Selection View Go Run Terminal Help

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- .github
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    - OrderServiceTest.class
- .gitignore
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- Jenkinsfile M
- pom.xml
- README.md

deployment.yaml 1 | Jenkinsfile M

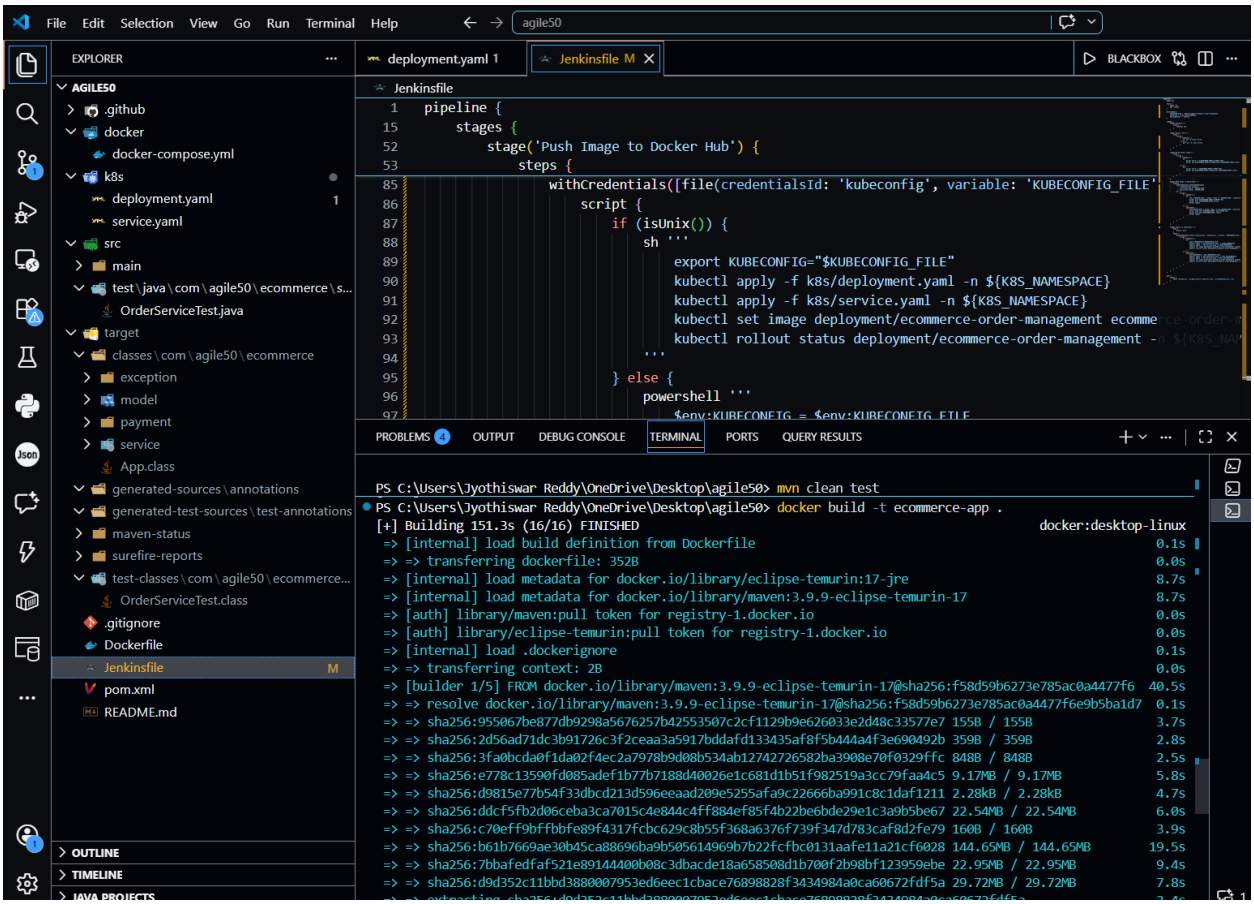
BLACKBOX

Jenkinsfile

```
1   pipeline {
15      stages {
52          stage('Push Image to Docker Hub') {
53              steps {
85                  withCredentials([file(credentialsId: 'kubeconfig', variable: 'KUBECONFIG_FILE'
86                      script {
87                          if (isUnix()) {
88                              sh '''
89                                  export KUBECONFIG="$KUBECONFIG_FILE"
90                                  kubectl apply -f k8s/deployment.yaml -n ${K8S_NAMESPACE}
91                                  kubectl apply -f k8s/service.yaml -n ${K8S_NAMESPACE}
92                                  kubectl set image deployment/ecommerce-order-management ecomme
93                                  kubectl rollout status deployment/ecommerce-order-management -n ${K8S_NAM
94                              '''
95                          } else {
96                              powershell '''
97                                  $env:KUBECONFIG = $env:KUBECONFIG_FILE
```

PROBLEMS 4 | OUTPUT | DEBUG CONSOLE | TERMINAL | PORTS | QUERY RESULTS

```
PS C:\Users\Jyothiswar Reddy\OneDrive\Desktop\agile50> mvn clean test
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running com.agile50.ecommerce.service.OrderServiceTest
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.090 s -- in com.agile50.ecommerce.servi
ce.OrderServiceTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
```



<u>Pipe line script:</u>

```
pipeline {

  agent any


  tools {

    maven 'M3'

    jdk 'Java21'

  }
```

```
environment {
   IMAGE_NAME = "ecommerce-order-management"
   DOCKERHUB_USER = " dhanushreddy17"
   IMAGE_TAG = "latest"
}

stages {

   stage('Checkout') {
      steps {
         git branch: 'main', url: 'https://github.com/DhanushReddy1717/agile_fat_room.git'
      }
   }

   stage('Build') {
      steps {
         bat 'mvn clean compile'
      }
   }

   stage('Test') {
      steps {
         bat 'mvn test'
      }
   }
```

```
stage('Package') {
   steps {
      bat 'mvn package'
   }
}

stage('Docker Build') {
   steps {
      bat "docker build -t %DOCKERHUB_USER%/%IMAGE_NAME%:%IMAGE_TAG% ."
   }
}

stage('Docker Login') {
   steps {
      withCredentials([usernamePassword(
         credentialsId: 'docker',
         usernameVariable: 'USERNAME',
         passwordVariable: 'PASSWORD'
      )]) {
         bat "docker login -u %USERNAME% -p %PASSWORD%"
      }
   }
}

stage('Docker Push') {
```

```
        steps {
            bat "docker push %DOCKERHUB_USER%/%IMAGE_NAME%:%IMAGE_TAG%"
        }
    }

    stage('Kubernetes Deploy') {
  steps {
    withCredentials([file(credentialsId: 'kubeconfig', variable: 'KUBECONFIG')]) {
        bat "kubectl --kubeconfig=%KUBECONFIG% apply -f deployment.yaml"
        bat "kubectl --kubeconfig=%KUBECONFIG% apply -f service.yaml"
    }
  }
}
  }
}
```

<u>Pipe line over view</u>

Jenkins / agiledhanush / #5 / Stages

#5

Rerun

Started by Jyotheswar reddy cp · Started 22 sec ago · Queued 19 ms · Build has been executing for 22 sec on built-in

Graph

Start → Tool Install → Checkout → Build → Test → Package → Docker Build → Docker Login → Docker Push → Kubernetes Deploy → End

Search

- Tool Install 0.36s
- Checkout 3s
- Build 8s
- Test 14s
- Package 10s
- Docker Build 5s
- Docker Login 4s
- Docker Push 8s
- Kubernetes Deploy 1s

Docker Build — 5s · Started 7s ago · Jenkins

- Use a tool from a predefined Tool Installation M3 — 0.1s
- Fetches the environment variables for a given tool in a list of 'FOO=bar' strings suitable for the withEnv step. — 0.14s
- Use a tool from a predefined Tool Installation Java21 — 87ms
- Fetches the environment variables for a given tool in a list of 'FOO=bar' strings suitable for the withEnv step. — 0.15s
- Windows Batch Script `docker build -t %DOCKERHUB_USER%/%IMAGE_NAME%:%IMAGE_TAG% .` — 4s

```
0
1 C:\ProgramData\Jenkins\.jenkins\workspace\agiledhanush>docker build -t dhanushreddy17/ecommerce-order-management:latest .
2 #0 building with "default" instance using docker driver
3
4 #1 [internal] load build definition from Dockerfile
5 #1 transferring dockerfile: 352B 0.0s done
6 #1 DONE 0.0s
7
8 #2 [auth] library/eclipse-temurin:pull token for registry-1.docker.io
9 #2 DONE 0.0s
```